Artificial Intelligence: An introduction

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Universidad Industrial de Santander

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Universidad Industrial de Santander

MACV
Motion Analysis and Computer Vision

BIVL^{2}ab
Biomedical Imaging, Vision and Learning Laboratory

Información del profesor

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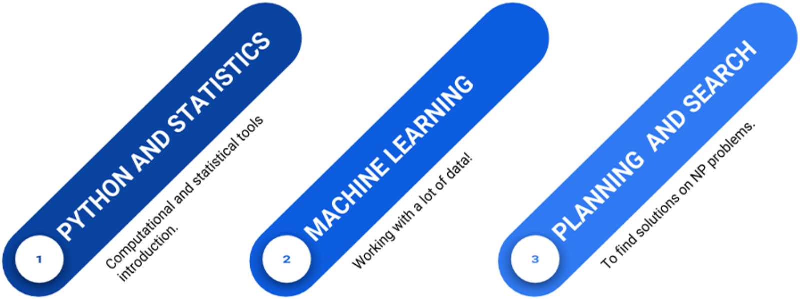
- Contenido del curso: `https://gitlab.com/bivl2ab/academico/cursos-uis/ai/ai-uis-student`

Course outline



- **The course** is mainly based on the **coding of I.A concepts** and the **development of small applications**
- **The course** is focused on the analysis and study of **I.A algorithms**

I.A. Course

Course

It is about learning the fundamental tools to face problems and challenges of I.A. as well as some computational tools and theoretical concepts.

First module: Python and statistics introduction

- Introduction to Python
- Introduction to statistics
- Introduction to Bayesian statistics

Second Module: Machine Learning

- Naive Bayes
- Classification trees
- Support Vector Machines
- Non-supervised learning
- Introduction to Deep-learning

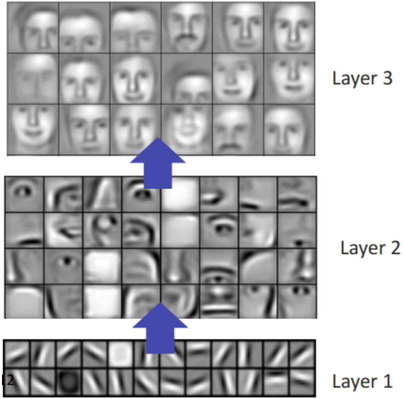
Third module: Planning and Searching

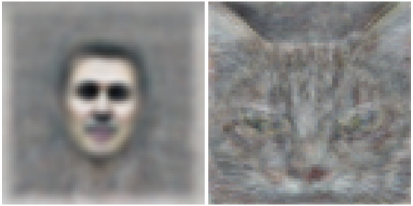
- Planning and searching
- Genetic algorithms (GA)
- Simulated annealing (SA)

Big challenges

Now, algorithms can create new things . . . as abstract as the art! (Netflix 5:15. Bill Nye)

Big challenges





Big challenges

Video 01

Big challenges

Video 02

ImageNet Challenge

IMAGENET

IMAGENET Netflix 22:30, Bill Nye

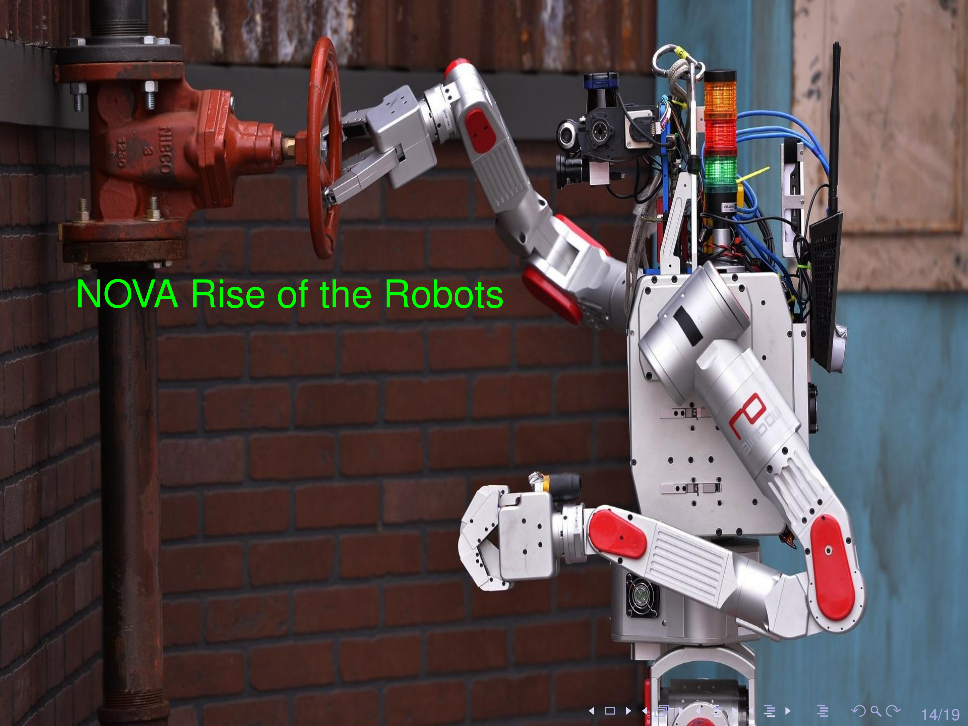
- 1,000 object classes (categories).
- Images:
  - 1.2 M train
  - 100k test.

2

As support for decisions making

- Medicine
- Politics
- Laws

NOVA Rise of the Robots



y ahora ...Atlas. Google again!

DARPA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RedBull
MDV
MDV

Structure of the Course

- 40% Talleres (Problemsets)
- 30% Parciales (Quizes)
- 30% + [10% ,20 %, 30 %] Proyecto funcional IA
  - 10%: Hasta + una unidad en una nota de un 10% (los parciales por ejemplo) o su equivalente en otros porcentajes.
  - Las notas o porcentajes adicionales se obtienen si fueron puntuales en todas las entregas.
- +10% Online courses (MOOC)
  - Esta vez, unicamente se aceptan cursos relacionados con I.A., machine learning, análisis de datos, o tecnologías relacionadas.
  - Solo se evalua el curso, si ha sido previamente aprobado antes del primer corte.

Curso en GITLAB:
`https://gitlab.com/bivl2ab/academico/cursos-uis/ai/ai-uis-student`

PROJECTS 2019-1

Proyectos realizados en 2019-1:

`https://gitlab.com/bivl2ab/academico/cursos-uis/ai/ai-uis-student/blob/master/projects/Proyectos_2019-1.md`

Thank you for your attention . . .

. . . It's time to wake up

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